

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457868.9186 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.144 +/- 0.024
Transit depth (Rp/Rs)^2	2.06 +/- 0.69 [%]
Orbital Inclination (inc)	83.59 +/- 2.29
Airmass coefficient 1 (a1)	1.025 +/- 0.015
Airmass coefficient 2 (a2)	-0.005 +/- 0.034
Scatter in the residuals of the lightcurve fit is	1.48 %
Best Comparison Star	#3 - [427, 218]
Optimal Aperture	5.27
Optimal Annulus	17.6
Transit Duration (day)	0.1022 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.144 ± 0.024 vs 0.10538 (deviation 1.48σ)
Duration fit vs published	0.1022 d vs 0.1167 d (deviation 1.57σ)
Mid-time O–C	1.4 min
Depth SNR	5.5
Residual scatter	1.48 %

Light curve

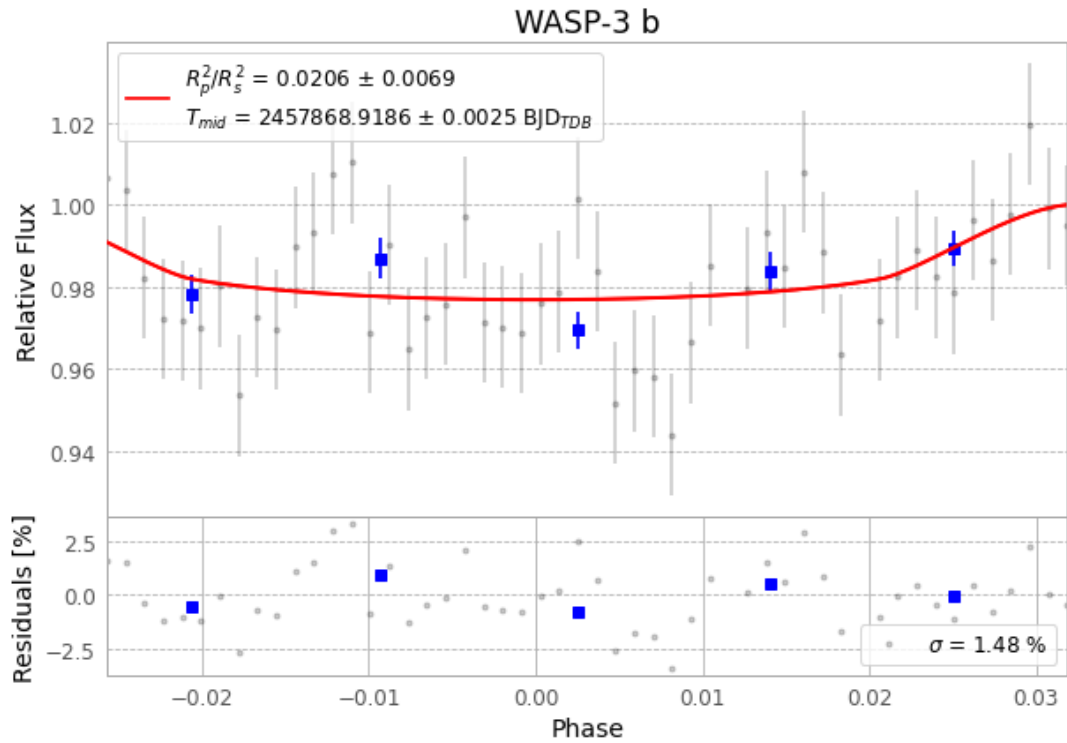


Figure 1 — FinalLightCurve_WASP-3 b_2017-04-25.png (SHA-256 ddeb2b6252b6b69f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [353, 268], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9445f205cb6b9aaf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit faa81f0

Data provenance

51 FITS frames (33 MB), WASP-3170425085112.FITS → WASP-3170425112412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-170425.log (c45203cced5d...), FinalLightCurve_WASP-3 b_2017-04-25.png (ddeb2b6252b6...), FinalLightCurve_WASP-3 b_2017-04-25.csv (1b5484b549f0...)

Compute resources

Wall time 685.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 15:45Z.